Sommative module 03

Every line of q1.py to q6.py, explained in simple English. Like pseudocode.

How to read this

- Grey line = the real code.
- Arrow line under it (→) = what the line do, in simple words.
- Each question start with: the idea, then the variables, then every line.
- The code is in French (the course is in French). The explanations are in English.

The notions (what each Python thing do)

variable	A box with a name. You put a value inside. `jours = 365` → box `jours`, put 365 inside. = mean « put inside », not « is equal ».
int	Whole number. No dot. `365`, `0`, `4`. For counting things.
float	Number with a dot. `4.00`, `273.15`, `5.2`. For money, temperature, hours.
str	Text. Always between quotes. `"oui"`, `"pierre"`.
bool	Only 2 values: `True` or `False`. A comparison like `montant < 50` give a bool.
list	Many values in one box, between `[ ]`. `["pierre", "papier", "ciseaux"]`.
#	Comment. Python skip it. It is a note for humans.
print()	Show something on the screen.
input()	Show a question, wait, user type something, press Enter. It always give back TEXT (str), even if user type a number.
int( )	Change text into whole number. `int("5")` → 5.
float( )	Change text into number with dot. `float("5.2")` → 5.2. `float("25")` → 25.0.
f"..."	f-string. Text where you can put a variable inside `{ }`. `f"Total : {total}"` → show « Total : » then the value of total.
:.2f	Inside `{ }`. Show the number with 2 digits after the dot. 4 → 4.00. `:.1f` = 1 digit. It only change the display, not the variable.
+ - * /	Plus, minus, times, divide. `/` always give a float (10 / 2 → 5.0).
< <= > >=	Smaller, smaller or equal, bigger, bigger or equal. Give True or False.
== vs =	== ask « is it equal? » (compare). = put a value in a box.
and	Both sides must be True.
or	At least one side must be True.
in / not in	`x in liste` → True if x is in the list. `not in` → True if x is NOT in the list.
+=	Add to what is already in the box. `compte1 += 1` = `compte1 = compte1 + 1`.

if	IF the condition is True → do the indented lines under it.
elif	ELSE IF. Only checked if the if (and elifs) before were False.
else	Everything else. Done only if all conditions before were False.
indentation	The 4 spaces at start of line. Tell Python: « this line is inside the if / loop above ».
for i in range(n)	Repeat n times. `i` count the turns: 0, 1, 2, … n-1.
while True	Repeat forever. Need `break` to get out.
break	Stop the loop now. Go to first line after the loop.
continue	Skip the rest of this turn. Go back to start of loop.
import	Bring a module (a toolbox) so we can use its tools.
random.randint(a, b)	Random whole number from a to b. a and b included. `randint(1, 4)` → 1, 2, 3 or 4.
random.choice(list)	Pick one random thing from the list.
math.ceil(x)	Round UP. 2.3 → 3. 5.2 → 6. 4.0 → 4.
.lower()	Make text small letters. `"OUI"` → `"oui"`.

One weak point (all 6 programs)

If user type letters when program want a number (example `abc`), `int()` or `float()` cannot change it. Program crash with `ValueError`. Not handled because assignment don't ask, and `try` / `except` is not seen in this module.

Question 1 — Hours, minutes and seconds in one year

The idea, in short

- Start with 365 days.
- Days × 24 = hours.
- Hours × 60 = minutes.
- Minutes × 60 = seconds.
- Show the 4 numbers.

The variables (the boxes)

Variable	Type	What it keep inside
jours	int	Number of days in one year. 365. The only number we give.
heures	int	Hours in one year. Calculated: 8760.
minutes	int	Minutes in one year. Calculated: 525600.
secondes	int	Seconds in one year. Calculated: 31536000.

Every line of q1.py

1 `# Question 1 : heures, minutes et secondes dans une année`

→ Comment. Python skip it. Title of the program: hours, minutes, seconds in a year.

2 *(empty line — only space, make code easy to read)*

3 `jours = 365`

→ Make box `jours`. Put 365 inside.

4 *(empty line — only space, make code easy to read)*

5 `heures = jours * 24       # 1 jour = 24 heures`

→ Take jours (365). Times 24 (hours in 1 day). Put answer in box `heures` → 8760. Text after # = comment, Python skip it.

6 `minutes = heures * 60     # 1 heure = 60 minutes`

→ Take heures. Times 60 (minutes in 1 hour). Put in box `minutes` → 525600.

7 `secondes = minutes * 60   # 1 minute = 60 secondes`

→ Take minutes. Times 60 (seconds in 1 minute). Put in box `secondes` → 31536000.

8 *(empty line — only space, make code easy to read)*

9 `print("Dans une année de 365 jours :")`

→ Show the text « Dans une année de 365 jours : ». Just text, no variable.

10 `print(f"Jours : {jours}")`

→ Show « Jours : » then the value of jours → « Jours : 365 ».

```
11  print(f"Heures : {heures}")
```

→ Show « Heures : » then the value of heures → « Heures : 8760 ».

```
12  print(f"Minutes : {minutes}")
```

→ Show « Minutes : » then the value of minutes.

```
13  print(f"Secondes : {secondes}")
```

→ Show « Secondes : » then the value of secondes.

If teacher ask…

Why not just write secondes = 31536000?

Assignment say: program must calculate. Also, if I change jours to 366, everything change alone.

What type is heures?

int. int times int = int.

Question 2 — Temperature conversion

The idea, in short

- Ask a temperature in Celsius.
- If it is colder than -273.15 → error, stop.
- Else → calculate Fahrenheit and Kelvin.
- Show the 3 with 2 digits after the dot.

The variables (the boxes)

Variable	Type	What it keep inside
celsius	float	Temperature the user type, in Celsius.
fahrenheit	float	Same temperature in Fahrenheit. Calculated.
kelvin	float	Same temperature in Kelvin. Calculated.

Every line of q2.py

1 `# Question 2 : conversion de température`

→ Comment. Python skip it. Title: temperature conversion.

2 *(empty line — only space, make code easy to read)*

3 `# float() accepte les nombres entiers et les nombres à virgule`

→ Comment. Note: `float()` accept whole numbers (25) and numbers with dot (25.5).

4 `celsius = float(input("Entrez une température en Celsius : "))`

→ Show question « Entrez une température en Celsius : ». User type. `input()` give text. `float()` change text to number. Put in box `celsius`.

5 *(empty line — only space, make code easy to read)*

6 `# -273.15 °C est la température la plus froide possible`

→ Comment. Note: -273.15 °C is the coldest possible (absolute zero).

7 `if celsius < -273.15:`

→ IF celsius is smaller than -273.15 → do line 8.

8 `    print("Erreur : la température ne peut pas être plus basse que -273.15 °C.")`

→ Show error message. Nothing else happen, program end.

9 `else:`

→ ELSE (temperature is OK) → do lines 10 to 13.

10 `    fahrenheit = (celsius * 9 / 5) + 32`

→ Fahrenheit = celsius times 9, divide by 5, plus 32. Put in box `fahrenheit`.

11 `    kelvin = celsius + 273.15`

→ Kelvin = celsius plus 273.15. Put in box `kelvin`.

12 *(empty line — only space, make code easy to read)*

13
```
    print(f"{celsius:.2f} °C = {fahrenheit:.2f} °F = {kelvin:.2f} K")
```
→ Show the 3 temperatures with their unit. `:.2f` = 2 digits after dot. 25 → « 25.00 °C = 77.00 °F = 298.15 K ».

If teacher ask…

Why float() and not int()?

Assignment say accept decimals. `int("25.5")` crash.

Why < and not <=?

Assignment say refuse temperature « smaller than » -273.15. So -273.15 itself is OK.

Why calculations inside the else?

If temperature is not possible, no reason to convert it.

Question 3 — Bill with discount and tax

The idea, in short

- Ask the price before tax.
- Choose discount: under 50 $ → 0 %, 50 to 100 $ → 5 %, over 100 $ → 10 %.
- Remove discount → sous-total.
- Tax 13 % on the sous-total (after discount).
- Show everything with 2 digits after dot.

The variables (the boxes)

Variable	Type	What it keep inside
montant	float	Price before tax and before discount. User type it.
rabais	int	Discount in percent: 0, 5 or 10.
economie	float	Money saved with the discount.
sous_total	float	Price after discount, before tax.
taxe	float	Tax (TVH 13 %) on the sous-total.
total	float	Final price to pay.

Every line of q3.py

1 `# Question 3 : facture avec rabais et taxes`

→ Comment. Python skip it. Title: bill with discount and taxes.

2 *(empty line — only space, make code easy to read)*

3 `montant = float(input("Montant de l'achat avant taxes : "))`

→ Ask price before tax. Change text to number with dot. Put in box `montant`.

4 *(empty line — only space, make code easy to read)*

5 `# On choisit le rabais selon le montant`

→ Comment. Note: now we choose the discount.

6 `if montant < 50:`

→ IF montant is smaller than 50 → do line 7.

7 `    rabais = 0`

→ Discount = 0 %. Put 0 in box `rabais`.

8 `elif montant <= 100:`

→ ELSE IF montant is 100 or smaller → do line 9. (We arrive here only if line 6 was False, so we know montant is 50 or more.)

9 `    rabais = 5`

→ Discount = 5 %. Put 5 in `rabais`.

10	`else:` → ELSE (montant is more than 100) → do line 11.
11	`    rabais = 10` → Discount = 10 %. Put 10 in `rabais`.
12	*(empty line — only space, make code easy to read)*
13	`economie = montant * rabais / 100` → Money saved = montant times rabais divide 100. Put in `economie`. Example: 120 × 10 / 100 = 12.
14	`sous_total = montant - economie` → Price after discount = montant minus economie. Put in `sous_total`. Example: 120 − 12 = 108.
15	`taxe = sous_total * 0.13          # TVH de 13 % sur le montant après rabais` → Tax = sous_total times 0.13. Tax is on price AFTER discount. Put in `taxe`. Example: 108 × 0.13 = 14.04.
16	`total = sous_total + taxe` → Final price = sous_total plus taxe. Put in `total`. Example: 108 + 14.04 = 122.04.
17	*(empty line — only space, make code easy to read)*
18	`print(f"Montant initial : {montant:.2f} $")` → Show « Montant initial : » + montant with 2 digits + « $ ».
19	`print(f"Rabais : {rabais} %")` → Show « Rabais : » + rabais + « % ». No `:.2f` because rabais is a whole number.
20	`print(f"Montant économisé : {economie:.2f} $")` → Show money saved, 2 digits.
21	`print(f"Sous-total après rabais : {sous_total:.2f} $")` → Show sous-total after discount, 2 digits.
22	`print(f"TVH (13 %) : {taxe:.2f} $")` → Show the tax, 2 digits.
23	`print(f"Total final : {total:.2f} $")` → Show the final total, 2 digits.

If teacher ask…

Discount for exactly 50 $? And 100 $?

Both 5 %. 50 < 50 is False → go to elif. 50 <= 100 is True. <= include 100.

Why elif and not 3 separate if?

With elif, Python stop at the first True one. Only one discount apply.

Why tax after discount?

Assignment say so. You pay tax on what you really pay.

Question 4 — Random draws statistics

The idea, in short

- Ask how many draws.
- Each draw: random number 1 to 4. Add 1 to the right counter.
- Show for each number: how many times + percent.
- Find and show the number that came most.

The variables (the boxes)

Variable	Type	What it keep inside
nombre_tirages	int	How many draws the user want.
compte1 … compte4	int	Counters. How many times 1, 2, 3, 4 came out. Start at 0.
i	int	Turn number of the for loop. Not used, only count the turns.
tirage	int	The random number of this turn (1, 2, 3 or 4).
plus_souvent	int	The number (1 to 4) that came the most. The winner.
plus_grand	int	The biggest counter found until now.

Every line of q4.py

1
```
# Question 4 : simulation de tirages aléatoires entre 1 et 4
```
→ Comment. Python skip it. Title: random draws between 1 and 4.

2 *(empty line — only space, make code easy to read)*

3
```
import random
```
→ Bring the `random` toolbox. Need it for random numbers.

4 *(empty line — only space, make code easy to read)*

5
```
nombre_tirages = int(input("Combien de tirages voulez-vous faire ? "))
```
→ Ask « Combien de tirages… ». Change text to whole number. Put in `nombre_tirages`.

6 *(empty line — only space, make code easy to read)*

7
```
# Un compteur pour chaque nombre
```
→ Comment. Note: one counter for each number.

8
```
compte1 = 0
```
→ Counter for number 1. Start at 0.

9
```
compte2 = 0
```
→ Counter for number 2. Start at 0.

10
```
compte3 = 0
```
→ Counter for number 3. Start at 0.

11 `compte4 = 0`

→ Counter for number 4. Start at 0.

12 *(empty line — only space, make code easy to read)*

13 `for i in range(nombre_tirages):`

→ Repeat lines 14 to 22, nombre_tirages times. Example: 1000 → repeat 1000 times.

14 `    tirage = random.randint(1, 4)`

→ Pick random number 1, 2, 3 or 4. Put in `tirage`.

15 `    if tirage == 1:`

→ IF tirage is 1 →

16 `        compte1 += 1`

→ add 1 to compte1.

17 `    elif tirage == 2:`

→ ELSE IF tirage is 2 →

18 `        compte2 += 1`

→ add 1 to compte2.

19 `    elif tirage == 3:`

→ ELSE IF tirage is 3 →

20 `        compte3 += 1`

→ add 1 to compte3.

21 `    else:`

→ ELSE (tirage can only be 4 now) →

22 `        compte4 += 1`

→ add 1 to compte4.

23 *(empty line — only space, make code easy to read)*

24 `if nombre_tirages > 0:`

→ Loop is finish. IF at least 1 draw → do lines 25 to 43. (Protection: cannot divide by 0 on line 25.)

25 `    print(f"1 : {compte1} fois – {compte1 / nombre_tirages * 100:.1f} %")`

→ Show « 1 : », how many times, « fois — », then percent. Percent = compte1 divide nombre_tirages times 100. `:.1f` = 1 digit after dot.

26 `    print(f"2 : {compte2} fois – {compte2 / nombre_tirages * 100:.1f} %")`

→ Same for number 2.

27 `    print(f"3 : {compte3} fois – {compte3 / nombre_tirages * 100:.1f} %")`

→ Same for number 3.

28 `    print(f"4 : {compte4} fois – {compte4 / nombre_tirages * 100:.1f} %")`

→ Same for number 4.

29 *(empty line — only space, make code easy to read)*

```
    # On cherche le nombre tiré le plus souvent
```
→ Comment. Note: now we find the number that came the most.

```
    plus_souvent = 1
```
→ Start: we say number 1 is the winner (for now).

```
    plus_grand = compte1
```
→ Remember compte1 as the biggest count (for now).

```
    if compte2 > plus_grand:
```
→ IF compte2 is bigger than the biggest until now →

```
        plus_souvent = 2
```
→ new winner is 2,

```
        plus_grand = compte2
```
→ new biggest count is compte2.

```
    if compte3 > plus_grand:
```
→ IF compte3 is bigger than the biggest until now →

```
        plus_souvent = 3
```
→ new winner is 3,

```
        plus_grand = compte3
```
→ new biggest count is compte3.

```
    if compte4 > plus_grand:
```
→ IF compte4 is bigger than the biggest until now →

```
        plus_souvent = 4
```
→ new winner is 4,

```
        plus_grand = compte4
```
→ new biggest count is compte4.

(empty line — only space, make code easy to read)

```
    print(f"Le nombre tiré le plus souvent : {plus_souvent}")
```
→ Show the winner number.

```
else:
```
→ ELSE (user asked 0 draws) →

```
    print("Aucun tirage effectué.")
```
→ Show « Aucun tirage effectué. » (no draw done).

If teacher ask…

Why if, if, if (lines 33, 36, 39) and not elif?

Need to check all 3. With elif, Python stop at first True and can miss a bigger one.

If two numbers have same count?

I use >, not >=. So first one found stay winner.

Why results change each time?

Random. More draws → each percent come closer to 25 %.

Why the if on line 24?

If 0 draws, line 25 divide by 0 → crash. The if stop that.

Question 5 — Rock, paper, scissors

The idea, in short

- Repeat: ask player choice.
- « quitter » → stop game.
- Bad choice → message, don't count, ask again.
- Computer pick random. Show both.
- Same → tie. Player beat computer → win. Else → lose.
- At the end: show wins, losses, ties, total.

The variables (the boxes)

Variable	Type	What it keep inside
choix_possibles	list	The 3 good choices: pierre, papier, ciseaux.
victoires	int	Number of wins. Start 0.
defaites	int	Number of losses. Start 0.
egalites	int	Number of ties. Start 0.
joueur	str	What the player type this turn (small letters).
ordinateur	str	What the computer pick this turn.

Every line of q5.py

1 `# Question 5 : jeu Pierre, papier, ciseaux`

→ Comment. Python skip it. Title: game rock, paper, scissors.

2 *(empty line — only space, make code easy to read)*

3 `import random`

→ Bring the `random` toolbox. For the computer choice.

4 *(empty line — only space, make code easy to read)*

5 `choix_possibles = ["pierre", "papier", "ciseaux"]`

→ Make a list with the 3 good choices. Put in `choix_possibles`.

6 *(empty line — only space, make code easy to read)*

7 `victoires = 0`

→ Wins counter. Start 0.

8 `defaites = 0`

→ Losses counter. Start 0.

9 `egalites = 0`

→ Ties counter. Start 0.

10 *(empty line — only space, make code easy to read)*

11
```
while True:
```
→ Repeat forever lines 12 to 36. Only way out: `break` (line 15).

12
```
    joueur = input("Pierre, papier ou ciseaux ? (quitter pour arrêter) : ").lower()
```
→ Ask player choice. `.lower()` make small letters (« Pierre » → « pierre »). Put in `joueur`.

13 *(empty line — only space, make code easy to read)*

14
```
    if joueur == "quitter":
```
→ IF player typed « quitter » →

15
```
        break
```
→ `break` = stop the loop. Jump to line 38.

16 *(empty line — only space, make code easy to read)*

17
```
    # Un choix invalide ne compte pas comme une partie
```
→ Comment. Note: bad choice don't count as a game.

18
```
    if joueur not in choix_possibles:
```
→ IF player choice is NOT in the list (example « banane ») →

19
```
        print("Choix invalide, essayez encore.")
```
→ Show « Choix invalide, essayez encore. »

20
```
        continue
```
→ `continue` = skip the rest, go back to line 12. No counter change.

21 *(empty line — only space, make code easy to read)*

22
```
    ordinateur = random.choice(choix_possibles)
```
→ Computer pick one random choice from the list. Put in `ordinateur`.

23
```
    print(f"Vous : {joueur} | Ordinateur : {ordinateur}")
```
→ Show player choice and computer choice.

24 *(empty line — only space, make code easy to read)*

25
```
    if joueur == ordinateur:
```
→ IF player and computer have same choice →

26
```
        print("Égalité !")
```
→ Show « Égalité ! »

27
```
        egalites += 1
```
→ Add 1 to egalites.

28
```
    # Pierre bat ciseaux, ciseaux battent papier, papier bat pierre
```
→ Comment. The rules: rock beat scissors, scissors beat paper, paper beat rock.

29
```
    elif (joueur == "pierre" and ordinateur == "ciseaux"
```
→ ELSE IF (player pierre AND computer ciseaux)

```
30          or joueur == "ciseaux" and ordinateur == "papier"
```
→ OR (player ciseaux AND computer papier)

```
31          or joueur == "papier" and ordinateur == "pierre"):
```
→ OR (player papier AND computer pierre) → player win.

```
32        print("Vous gagnez !")
```
→ Show « Vous gagnez ! »

```
33        victoires += 1
```
→ Add 1 to victoires.

```
34    else:
```
→ ELSE (not tie, not win → it is a loss) →

```
35        print("Vous perdez.")
```
→ Show « Vous perdez. »

```
36        defaites += 1
```
→ Add 1 to defaites. Then loop go back to line 12.

37 *(empty line — only space, make code easy to read)*

```
38 print()
```
→ Loop is finish (player quit). `print()` empty = show one empty line.

```
39 print(f"Victoires : {victoires}")
```
→ Show number of wins.

```
40 print(f"Défaites : {defaites}")
```
→ Show number of losses.

```
41 print(f"Égalités : {egalites}")
```
→ Show number of ties.

```
42 print(f"Total de parties : {victoires + defaites + egalites}")
```
→ Show total games = wins + losses + ties (added right here).

If teacher ask…

break vs continue?

`break` = leave the loop, game over. `continue` = skip this turn, start next turn.

Why bad choice not counted?

`continue` go back to start BEFORE the lines that add to counters.

Why while and not for?

We don't know how many games player want.

And + or on lines 29–31, how Python read it?

`and` is done first. So each line is one pair. The 3 pairs are joined with `or`.

Question 6 — Parking price

The idea, in short

- Ask: electric car? and how many hours?
- Round hours UP (2.3 h → 3 h). Minimum 1 hour.
- Price: 4.00 $ first hour + 2.50 $ each other hour.
- Max 20.00 $.
- Electric → 20 % discount.
- Show everything.

The variables (the boxes)

Variable	Type	What it keep inside
electrique	str	Answer « oui » or « non » (small letters).
duree	float	Hours the user type. Can have dot (5.2).
heures_facturees	int	Hours we charge: duree rounded up. Minimum 1.
prix	float	Price before discount. Max 20.00.
rabais	float	Discount money. 20 % of prix if electric, else 0.
total	float	Final price to pay.

Every line of q6.py

1 `# Question 6 : frais de stationnement`
→ Comment. Python skip it. Title: parking price.

2 *(empty line — only space, make code easy to read)*

3 `import math`
→ Bring the `math` toolbox. For `math.ceil` (round up).

4 *(empty line — only space, make code easy to read)*

5 `electrique = input("Le véhicule est-il électrique ? (oui/non) : ").lower()`
→ Ask « electric? (oui/non) ». `.lower()` small letters. Put text in `electrique`.

6 `duree = float(input("Nombre d'heures de stationnement : "))`
→ Ask how many hours. Change text to number with dot. Put in `duree`.

7 *(empty line — only space, make code easy to read)*

8 `# math.ceil arrondit vers le haut : 2.3 h devient 3 h`
→ Comment. Note: math.ceil round up, 2.3 h become 3 h.

9 `heures_facturees = math.ceil(duree)`
→ Round duree UP. 5.2 → 6. Put in `heures_facturees`.

10 *(empty line — only space, make code easy to read)*

11
```
# Il faut payer au moins 1 heure
```
→ Comment. Note: must pay at least 1 hour.

12
```
if heures_facturees < 1:
```
→ IF heures_facturees smaller than 1 (user typed 0) →

13
```
    heures_facturees = 1
```
→ heures_facturees = 1. Pay minimum 1 hour.

14 *(empty line — only space, make code easy to read)*

15
```
# Première heure : 4.00 $, chaque heure en plus : 2.50 $
```
→ Comment. Note: first hour 4.00 $, each other hour 2.50 $.

16
```
prix = 4.00 + (heures_facturees - 1) * 2.50
```
→ Price = 4.00 + (hours − 1) × 2.50. Minus 1 because first hour already in the 4.00. Put in `prix`. Example 6 h: 4 + 5 × 2.50 = 16.50.

17 *(empty line — only space, make code easy to read)*

18
```
# Le prix ne peut pas dépasser 20.00 $
```
→ Comment. Note: price cannot go over 20.00 $.

19
```
if prix > 20.00:
```
→ IF prix more than 20 →

20
```
    prix = 20.00
```
→ prix = 20. Daily max.

21 *(empty line — only space, make code easy to read)*

22
```
# Rabais de 20 % pour un véhicule électrique
```
→ Comment. Note: 20 % discount for electric car.

23
```
if electrique == "oui":
```
→ IF electrique is « oui » →

24
```
    rabais = prix * 0.20
```
→ Discount = prix × 0.20. Put in `rabais`. Example: 16.50 × 0.20 = 3.30.

25
```
else:
```
→ ELSE (not electric) →

26
```
    rabais = 0.00
```
→ Discount = 0.

27 *(empty line — only space, make code easy to read)*

28
```
total = prix - rabais
```
→ Final price = prix minus rabais. Put in `total`. Example: 16.50 − 3.30 = 13.20.

29 *(empty line — only space, make code easy to read)*

```
30 print(f"Durée : {duree} heures")
```
→ Show the duration typed.

```
31 print(f"Heures facturées : {heures_facturees}")
```
→ Show the hours charged.

```
32 print(f"Prix avant rabais : {prix:.2f} $")
```
→ Show price before discount, 2 digits.

```
33 print(f"Rabais véhicule électrique : {rabais:.2f} $")
```
→ Show discount, 2 digits.

```
34 print(f"Total : {total:.2f} $")
```
→ Show final total, 2 digits.

If teacher ask…

Why math.ceil and not round()?

`round(2.3)` → 2. We need 3. Started hour = full hour.

Why hours – 1?

First hour is already paid in the 4.00 $. The 2.50 $ is only for the next hours.

When the 20 $ max start?

7 h = 19.00 $. 8 h would be 21.50 $ → 20.00 $. So from 8 hours.

Why lines 12–13?

If user type 0, price would be 1.50 $. Not logic. Now minimum = 1 hour = 4.00 $.

Max before or after discount?

Before. Max is parking price. Discount come after. Electric car pay max 16.00 $.